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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (CSE) (CBCS) Degree Examinations
May-June 2025

23CSE201 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| Unit – I | | | | |
| 1. a) What is Intelligence? Elaborate on the key components that create Artificial Intelligence. | 10 | L*1 | 1 | 1 |
| b) What is an uninformed search in AI and how does it differ from informed search? What are the two main types of uninformed search, and how do they function? Explain with an example. | 10 | L2 | 1 | 1 |
| 2. a) Provide a list of any five application areas of artificial intelligence and explain each in detail. | 10 | L2 | 1 | 1 |
| b) What is the Hill Climbing algorithm and what are its main drawbacks? How do these drawbacks impact the algorithm's ability to find an optimal solution? | 10 | L2 | 1 | 1 |
| Unit – II | | | | |
| 3. a) What is a Knowledge based system? Explain procedural and declarative knowledge with an example. | 10 | L2 | 2 | 1 |
| b) What is an expert system in artificial intelligence and list key advantages and limitations of it. | 10 | L1 | 2 | 1 |
| 4. a) Explain each phase of the expert system life cycle in detail, highlighting the processes involved at every stage in the development of an expert system. | 10 | L2 | 2 | 1 |
| b) Describe rule-based expert system architecture with a neat diagram. | 10 | L2 | 2 | 2 |
| Unit – III | | | | |
| 5. a) Discuss Supervised Learning, Unsupervised Learning, and Reinforcement Learning with suitable examples. | 10 | L1 | 3 | 1 |
| b) Consider a small dataset for predicting whether a student will pass or fail based on their study hours and number of assignments completed. Apply K-NN algorithm and classify the new student H with Study Hours = 3 and Assignments Completed = 4 using K=3. | | | | |

Student	Study Hours (X)	Assignments Completed (Y)	Result
A	1	2	Fail
B	3	2	Fail
C	3	1	Pass
D	4	5	Pass
E	5	5	Pass
F	6	4	Pass
G	1	1	Fail

	10	L3	3	1
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P.T.O.

6. a) Discuss the process of designing learning systems. What are the key challenges involved in this process? Explain. 10 L2 3 1
- b) Discuss the Support Vector Machine (SVM) algorithm. Outline its key concept, working mechanism, and the advantages and disadvantages of using SVM. 10 L2 3 1

Unit – IV

7. a) What is an Artificial Neural Network? Explain the working of backpropagation neural networks with diagram. 10 L2 4 1
- b) Compare and contrast regression and classification. Provide examples where each type of model is most appropriate. 10 L2 4 1
8. a) Describe the design of the ALVINN system for autonomous vehicle steering and write the characteristics of appropriate problems for neural network learning. 10 L2 4 1
- b) Define the following terms and explain their significance of Confusion matrix, Precision, Recall, F1 Score and AUC. Explain each of these metrics with examples. 10 L2 4 2

Unit – V

9. a) Describe the K-means clustering algorithm and its working principle with an example. 10 L2 5 1
- b) How does the Random Forest algorithm work? Illustrate it. 10 L2 5 2
10. a) Explain the concept of ensemble learning and elaborate two main ensemble strategies, boosting and bagging. 10 L2 5 1
- b) What is Reinforcement Learning? Explain the Reinforcement learning with a neat diagram. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
